UNIT - V

Fourier Series & Fourier Transforms

(Euler coefficients, even/odd, arbitrary interval, half-range, Fourier integral, sine/cosine transforms)

PART A : FOURIER SERIES

1. Introduction & Dirichlet's Conditions

A function $f(x)$ is PERIODIC with period T if $f(x+T)=f(x)$. A periodic function of period 2π can be represented as a series of sines and cosines:

$$f(x)=\frac{a_0}{2}+\sum_{n=1}^{\infty}(a_n\cos nx+b_n\sin nx)$$

This is the FOURIER SERIES of $f(x)$; a_0, a_n, b_n are FOURIER COEFFICIENTS.

Dirichlet's conditions (sufficient for the series to exist and converge)

1. *$f(x)$ is single-valued, finite and periodic in the interval $(c, c+2\pi)$.*
2. *$f(x)$ has a finite number of discontinuities (finite jumps) in one period.*
3. *$f(x)$ has a finite number of maxima and minima in one period.*

Convergence

Then the Fourier series converges: to $f(x)$ at every point of CONTINUITY, and

$$\text{to } \frac{1}{2}[f(x^-)+f(x^+)] \text{ at a point of DISCONTINUITY}$$

At end points $x=c$ and $x=c+2\pi$ the sum is $\frac{1}{2}[f(c)+f(c+2\pi)]$.

2. Euler's Formulae for the Coefficients

Orthogonality over a period of length 2π gives (for m, n positive integers):

$$\int_c^{c+2\pi} \sin nx\,dx = \int_c^{c+2\pi} \cos nx\,dx = 0, \qquad \int \sin mx \cos nx\,dx = 0$$

$$\int_c^{c+2\pi} \sin mx \sin nx\,dx = \int_c^{c+2\pi} \cos mx \cos nx\,dx = 0 \text{ for } m \neq n \text{ and } = \pi$$

for $m = n$.

Euler's formulae (interval c to c+2pi)

$$a_0 = \frac{1}{\pi}\int_c^{c+2\pi} f(x)\,dx$$

$$a_n = \frac{1}{\pi}\int_c^{c+2\pi} f(x)\cos nx\,dx, \qquad b_n = \frac{1}{\pi}\int_c^{c+2\pi} f(x)\sin nx\,dx$$

- *Most common intervals: $(0, 2\pi)$ with $c = 0$ and $(-\pi, \pi)$ with $c = -\pi$.*
- *Multiply $f(x) = \frac{a_0}{2} + \sum \ldots$ by $\cos nx$ (or $\sin nx$), integrate over a period; only one term survives - that gives the formulae.*

Useful tools for integration

Bernoulli's generalised integration by parts:

$\int u\,v\,dx = uv_1 - u'v_2 + u''v_3 - \cdots$ *where $v_1, v_2, \ldots$ are successive integrals of v and $u', u'', \ldots$ successive derivatives of u (stops when $u^{(k)} = 0$).*

n **integer**	$\sin n\pi$	$\cos n\pi$	$\sin 2n\pi$	$\cos 2n\pi$	$\sin \frac{n\pi}{2}$
value	0	$(-1)^n$	0	1	$0, 1, 0, -1, \ldots$

Also $\int e^{ax} \sin bx\,dx = \frac{e^{ax}}{a^2 + b^2}(a\sin bx - b\cos bx)$,

$\int e^{ax} \cos bx\,dx = \frac{e^{ax}}{a^2 + b^2}(a\cos bx + b\sin bx)$.

Example 1. Expand $f(x)=x^2$ in the interval $(0,2\pi)$

$$a_0=\frac{1}{\pi}\int_0^{2\pi}x^2dx=\frac{8\pi^2}{3},\ a_n=\frac{1}{\pi}\int_0^{2\pi}x^2\cos nx\,dx=\frac{4}{n^2},$$
$$b_n=\frac{1}{\pi}\int_0^{2\pi}x^2\sin nx\,dx=-\frac{4\pi}{n}$$

$$x^2=\frac{4\pi^2}{3}+\sum_{n=1}^{\infty}\left[\frac{4}{n^2}\cos nx-\frac{4\pi}{n}\sin nx\right],\quad 0<x<2\pi$$

At $x=0$ (end point) the sum is $\frac{1}{2}[f(0)+f(2\pi)]=2\pi^2$ so

$$2\pi^2=\tfrac{4\pi^2}{3}+4\textstyle\sum\frac{1}{n^2}\ \Rightarrow\ \sum\frac{1}{n^2}=\frac{\pi^2}{6}.$$

Example 2. Expand $f(x)=e^{bx}$ in $(0,2\pi)$

With $K=e^{2\pi b}-1$: $a_0=\frac{K}{\pi b},\ a_n=\frac{b\,K}{\pi(b^2+n^2)},\ b_n=-\frac{n\,K}{\pi(b^2+n^2)}$

$$e^{bx}=\frac{K}{\pi}\left[\frac{1}{2b}+\sum_{n=1}^{\infty}\frac{b\cos nx-n\sin nx}{b^2+n^2}\right]$$

Example 3. $f(x)=-\pi$ for $-\pi<x<0$; $f(x)=x$ for $0<x<\pi$

$$a_0=\frac{1}{\pi}\left[\int_{-\pi}^{0}(-\pi)dx+\int_0^{\pi}x\,dx\right]=\frac{1}{\pi}\left[-\pi^2+\frac{\pi^2}{2}\right]=-\frac{\pi}{2}$$
$$a_n=\frac{1}{\pi}\left[0+\int_0^{\pi}x\cos nx\,dx\right]=\frac{(-1)^n-1}{\pi n^2},$$
$$b_n=\frac{1}{\pi}\left[-\pi\frac{(-1)^n-1}{n}-\frac{\pi(-1)^n}{n}\right]=\frac{1-2(-1)^n}{n}$$

$$f(x)=-\frac{\pi}{4}+\sum_{n=1}^{\infty}\left[\frac{(-1)^n-1}{\pi n^2}\cos nx+\frac{1-2(-1)^n}{n}\sin nx\right]$$

At $x=0$ the sum $=\frac{1}{2}[f(0^-)+f(0^+)]=\frac{1}{2}(-\pi+0)=-\frac{\pi}{2}$ (point of discontinuity).

3. Even and Odd Functions

f is EVEN if $f(-x)=f(x)$ (symmetric about the y-axis), ODD if $f(-x)=-f(x)$. For the interval $(-\pi,\pi)$:

	Even function	Odd function
a_0	$\frac{2}{\pi}\int_0^{\pi} f\,dx$	0
a_n	$\frac{2}{\pi}\int_0^{\pi} f\cos nx\,dx$	0
b_n	0	$\frac{2}{\pi}\int_0^{\pi} f\sin nx\,dx$
Series	*COSINE series only*	*SINE series only*

- *Products:* even×even = even, odd×odd = even, even×odd = odd.

$\int_{-a}^{a}$ *of an odd function* $= 0$; *of an even function* $= 2\int_0^a$.

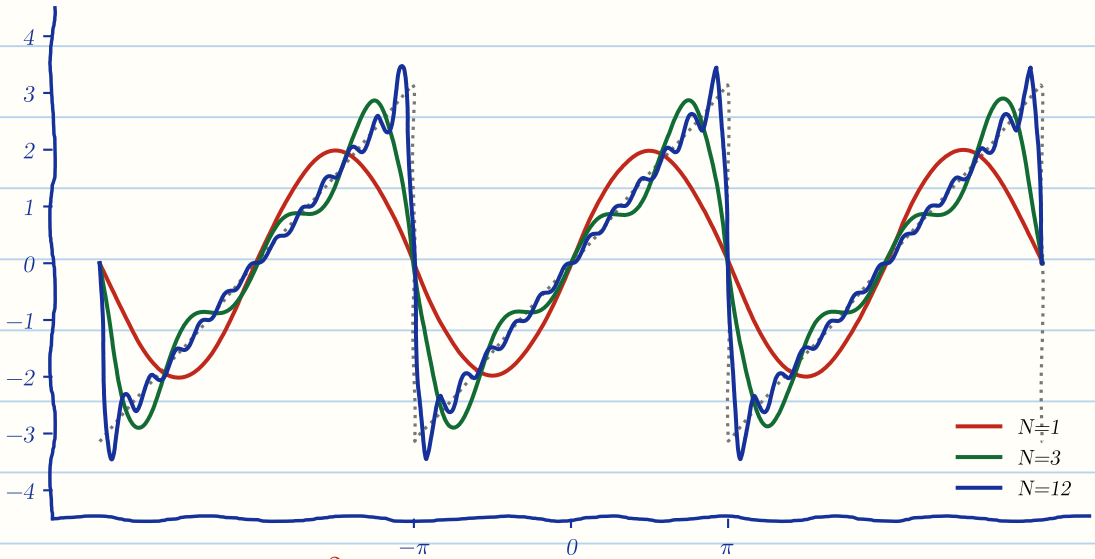
Example 4. $f(x) = x$ **in** $(-\pi, \pi)$ **(odd)**

$$b_n = \frac{2}{\pi}\int_0^{\pi} x\sin nx\,dx = \frac{2}{\pi}\left[-\frac{x\cos nx}{n} + \frac{\sin nx}{n^2}\right]_0^{\pi} = \frac{2(-1)^{n+1}}{n}$$

$$x = 2\left[\sin x - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} - \cdots\right]$$

Putting $x = \frac{\pi}{2}$: $\frac{\pi}{2} = 2\left(1 - \frac{1}{3} + \frac{1}{5} - \cdots\right) \Rightarrow \frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \cdots$ *(Leibniz series).*

Partial sums of the Fourier series of $f(x) = x$



Example 5. $f(x) = x^2$ **in** $(-\pi, \pi)$ **(even)**

$$a_0 = \frac{2}{\pi}\int_0^{\pi} x^2 dx = \frac{2\pi^2}{3},\quad a_n = \frac{2}{\pi}\int_0^{\pi} x^2\cos nx\,dx = \frac{4(-1)^n}{n^2},\quad b_n = 0$$

$$x^2 = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}\cos nx = \frac{\pi^2}{3} - 4\cos x + \cos 2x - \frac{4}{9}\cos 3x + \cdots$$

At $x=\pi$: $\pi^2 = \frac{\pi^2}{3} + 4\sum\frac{1}{n^2} \Rightarrow \sum\frac{1}{n^2} = \frac{\pi^2}{6}$; *at* $x=0$:

$\sum\frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}$.

Example 6. $f(x)=|x|$ **in** $(-\pi,\pi)$ **(even)**

$a_0 = \frac{2}{\pi}\int_0^{\pi} x\,dx = \pi,\ a_n = \frac{2}{\pi}\int_0^{\pi} x\cos nx\,dx = \frac{2[(-1)^n-1]}{\pi n^2}$ (*so* $a_n = -\frac{4}{\pi n^2}$ *for odd* n, 0 *for even* n)

$$|x| = \frac{\pi}{2} - \frac{4}{\pi}\left[\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \cdots\right]$$

At $x=0$: $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots = \frac{\pi^2}{8}$.

Example 7. $f(x)=x^3$ **in** $(-\pi,\pi)$ **(odd)**

$b_n = \frac{2}{\pi}\int_0^{\pi} x^3\sin nx\,dx = 2(-1)^{n+1}\left[\frac{\pi^2}{n} - \frac{6}{n^3}\right] \Rightarrow$

$$x^3 = \sum_{n=1}^{\infty} 2(-1)^{n+1}\left[\frac{\pi^2}{n} - \frac{6}{n^3}\right]\sin nx$$

4. Fourier Series in an Arbitrary Interval

For a function of period $2l$ *defined in* $(c, c+2l)$ *put* $x \to \frac{\pi x}{l}$:

Interval (c, c+2l)

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n\cos\frac{n\pi x}{l} + b_n\sin\frac{n\pi x}{l}\right)$$

$$a_0 = \frac{1}{l}\int_c^{c+2l} f\,dx,\quad a_n = \frac{1}{l}\int_c^{c+2l} f\cos\frac{n\pi x}{l}\,dx,\quad b_n = \frac{1}{l}\int_c^{c+2l} f\sin\frac{n\pi x}{l}\,dx$$

- *On* $(-l,l)$: *even* $f \Rightarrow a_0 = \frac{2}{l}\int_0^{l} f,\ a_n = \frac{2}{l}\int_0^{l} f\cos\frac{n\pi x}{l},\ b_n = 0$; *odd* f $\Rightarrow b_n = \frac{2}{l}\int_0^{l} f\sin\frac{n\pi x}{l}$.

Example 8. $f(x)=x^2$ **in** $(-l,l)$

$$\text{Even: } a_0=\frac{2}{l}\int_0^l x^2dx=\frac{2l^2}{3},\quad a_n=\frac{2}{l}\int_0^l x^2\cos\frac{n\pi x}{l}dx=\frac{4l^2(-1)^n}{n^2\pi^2}$$

$$x^2=\frac{l^2}{3}+\frac{4l^2}{\pi^2}\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}\cos\frac{n\pi x}{l}$$

Example 9. $f(x)=2x-x^2$ **in** $(0,2)$ $(c=0,\ l=1)$

$$a_0=\int_0^2(2x-x^2)dx=\frac{4}{3},\quad a_n=\int_0^2(2x-x^2)\cos n\pi x\,dx=-\frac{4}{n^2\pi^2},$$

$$b_n=\int_0^2(2x-x^2)\sin n\pi x\,dx=0$$

$$2x-x^2=\frac{2}{3}-\frac{4}{\pi^2}\sum_{n=1}^{\infty}\frac{\cos n\pi x}{n^2}$$

(The function is symmetric about $x=1$, which is why all sine terms vanish.)

5. Half-Range Fourier Series

A function given only on $(0,l)$ can be expanded in a series of SINES only or COSINES only (extend it as an odd or even function to $(-l,l)$):

Half-range expansions on (0, l)

$$\text{Sine series: } f(x)=\sum_{n=1}^{\infty}b_n\sin\frac{n\pi x}{l},\qquad b_n=\frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}\,dx$$

$$\text{Cosine series: } f(x)=\frac{a_0}{2}+\sum_{n=1}^{\infty}a_n\cos\frac{n\pi x}{l}$$

$$a_0=\frac{2}{l}\int_0^l f\,dx,\qquad a_n=\frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}\,dx$$

Example 10. $f(x)=x$ **in** $(0,\pi)$ **: (a) cosine series (b) sine series**

$$(a)\ a_0=\frac{2}{\pi}\int_0^{\pi}x\,dx=\pi,\ a_n=\frac{2}{\pi}\int_0^{\pi}x\cos nx\,dx=\frac{2[(-1)^n-1]}{\pi n^2}:$$

$$x=\frac{\pi}{2}-\frac{4}{\pi}\left[\cos x+\frac{\cos 3x}{3^2}+\frac{\cos 5x}{5^2}+\cdots\right]$$

(b) $b_n = \frac{2}{\pi}\int_0^{\pi} x\sin nx\,dx = \frac{2(-1)^{n+1}}{n}$:

$$x = 2\left[\sin x - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} - \cdots\right]$$

Example 11. Sine series of $f(x) = x(\pi - x)$ in $(0, \pi)$

$b_n = \frac{2}{\pi}\int_0^{\pi} x(\pi - x)\sin nx\,dx = \frac{4[1-(-1)^n]}{\pi n^3}$ (*zero for even* n, $\frac{8}{\pi n^3}$ *for odd* n)

$$x(\pi - x) = \frac{8}{\pi}\left[\sin x + \frac{\sin 3x}{3^3} + \frac{\sin 5x}{5^3} + \cdots\right]$$

Example 12. Cosine series of $f(x) = \sin x$ in $(0, \pi)$

$a_0 = \frac{2}{\pi}\int_0^{\pi} \sin x\,dx = \frac{4}{\pi}$; $a_n = \frac{2}{\pi}\int_0^{\pi} \sin x\cos nx\,dx = \frac{2[1+(-1)^n]}{\pi(1-n^2)}$ $(n \neq 1)$, $a_1 = 0$.

$$\sin x = \frac{2}{\pi} - \frac{4}{\pi}\left[\frac{\cos 2x}{1\cdot 3} + \frac{\cos 4x}{3\cdot 5} + \frac{\cos 6x}{5\cdot 7} + \cdots\right]$$

Parseval's identity (extra)

$$\frac{1}{\pi}\int_{-\pi}^{\pi}[f(x)]^2 dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty}(a_n^2 + b_n^2)$$

PART B : FOURIER TRANSFORMS

6. Fourier Integral Theorem (without proof)

A NON-periodic function can be seen as the limit of a Fourier series when the period $2l \to \infty$. *If* $f(x)$ *satisfies Dirichlet's conditions in every finite interval and* $\int_{-\infty}^{\infty}|f(x)|dx$ *converges, then*

Fourier integral theorem

$$f(x)=\frac{1}{\pi}\int_0^\infty\int_{-\infty}^\infty f(t)\cos\lambda(t-x)\,dt\,d\lambda$$

At a point of discontinuity the left side is replaced by $\frac{1}{2}[f(x^-)+f(x^+)]$.

Fourier sine and cosine integrals

$$f(x)=\frac{2}{\pi}\int_0^\infty \cos\lambda x\left[\int_0^\infty f(t)\cos\lambda t\,dt\right]d\lambda \quad (f \text{ even})$$

$$f(x)=\frac{2}{\pi}\int_0^\infty \sin\lambda x\left[\int_0^\infty f(t)\sin\lambda t\,dt\right]d\lambda \quad (f \text{ odd})$$

Complex form of the Fourier integral

$$f(x)=\frac{1}{2\pi}\int_{-\infty}^\infty e^{-i\lambda x}\left[\int_{-\infty}^\infty f(t)\,e^{i\lambda t}dt\right]d\lambda$$

Example 13. $f(x)=1$ **for** $|x|<1$, 0 **for** $|x|>1$ **: express as a Fourier integral and deduce** $\displaystyle\int_0^\infty \frac{\sin\lambda}{\lambda}d\lambda$

f *is even.* $\displaystyle\int_0^\infty f(t)\cos\lambda t\,dt=\int_0^1\cos\lambda t\,dt=\frac{\sin\lambda}{\lambda}.$

$$f(x)=\frac{2}{\pi}\int_0^\infty\frac{\sin\lambda\cos\lambda x}{\lambda}\,d\lambda$$

Put $x=0$ $(f(0)=1)$: $\displaystyle\frac{2}{\pi}\int_0^\infty\frac{\sin\lambda}{\lambda}d\lambda=1 \;\Rightarrow\; \int_0^\infty\frac{\sin\lambda}{\lambda}d\lambda=\frac{\pi}{2}.$

At $x=1$ *(jump): value* $\frac{1}{2}$*, giving* $\displaystyle\int_0^\infty\frac{\sin\lambda\cos\lambda}{\lambda}d\lambda=\frac{\pi}{4}.$

Example 14. Fourier integral of $f(x)=e^{-ax}$ $(x>0)$**, even extension** $e^{-a|x|}$

$$\int_0^\infty e^{-at}\cos\lambda t\,dt=\frac{a}{a^2+\lambda^2}\;\Rightarrow$$

$$e^{-a|x|} = \frac{2}{\pi}\int_0^\infty \frac{a\cos\lambda x}{a^2+\lambda^2}\,d\lambda \;\Rightarrow\; \int_0^\infty \frac{\cos\lambda x}{a^2+\lambda^2}\,d\lambda = \frac{\pi}{2a}e^{-a|x|}$$

7. Fourier Transform, Sine & Cosine Transforms

Fourier transform pair

$$F(s) = F\{f(x)\} = \int_{-\infty}^{\infty} f(x)\,e^{isx}\,dx$$

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(s)\,e^{-isx}\,ds \qquad \text{(inverse transform)}$$

Cosine & Sine transforms (and their inverses)

$$F_c(s) = \int_0^\infty f(x)\cos sx\,dx, \qquad f(x) = \frac{2}{\pi}\int_0^\infty F_c(s)\cos sx\,ds$$

$$F_s(s) = \int_0^\infty f(x)\sin sx\,dx, \qquad f(x) = \frac{2}{\pi}\int_0^\infty F_s(s)\sin sx\,ds$$

(Some books use the symmetric form with factor $1/\sqrt{2\pi}$ (or $\sqrt{2/\pi}$); always follow the definition used in your question.)

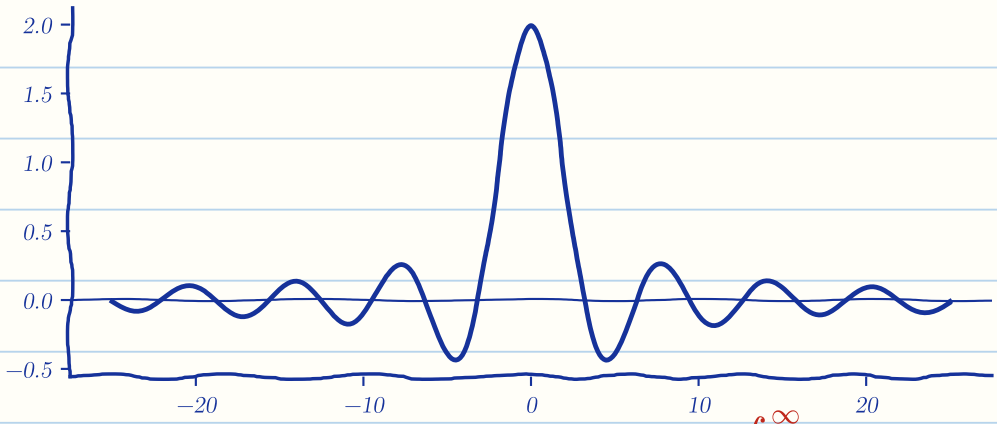
Standard transforms (a > 0)

$f(x)$	$F_c(s)$	$F_s(s)$
e^{-ax}	$\frac{a}{a^2+s^2}$	$\frac{s}{a^2+s^2}$
$1\ (0<x<a),\ 0\ (x>a)$	$\frac{\sin as}{s}$	$\frac{1-\cos as}{s}$
$x\,e^{-ax}$	$\frac{a^2-s^2}{(a^2+s^2)^2}$	$\frac{2as}{(a^2+s^2)^2}$
$\frac{e^{-ax}}{x}$	-	$\tan^{-1}\frac{s}{a}$
$e^{-a^2x^2}$	$\frac{1}{2a}\sqrt{\pi}\,e^{-s^2/4a^2}$	-

Example 15. Fourier transform of $f(x)=1$ for $|x|<a$, 0 for $|x|>a$

$$F(s)=\int_{-a}^{a} e^{isx}\,dx=\frac{e^{isa}-e^{-isa}}{is}=\frac{2\sin as}{s}\quad (a\ \text{"sinc" curve}).$$

Transform of the pulse $|x|<1$: $F(s)=2\sin s/s$



Example 16. Find F_s and F_c of e^{-ax}; hence $\displaystyle\int_0^\infty \frac{s\sin sx}{a^2+s^2}\,ds$

$F_s=\displaystyle\int_0^\infty e^{-ax}\sin sx\,dx=\frac{s}{a^2+s^2}$. *Inverse:* $e^{-ax}=\dfrac{2}{\pi}\displaystyle\int_0^\infty \frac{s\sin sx}{a^2+s^2}\,ds\ \Rightarrow$

$$\int_0^\infty \frac{s\sin sx}{a^2+s^2}\,ds=\frac{\pi}{2}e^{-ax}.$$

Example 17. Self-reciprocal: $f(x)=e^{-x^2/2}$

$F(s)=\displaystyle\int_{-\infty}^{\infty} e^{-x^2/2}e^{isx}\,dx=\sqrt{2\pi}\,e^{-s^2/2}$ *(complete the square:* $-\frac{1}{2}(x-is)^2-\frac{s^2}{2}$*).*

Example 18. Find $f(x)$ if $F_c(s)=\dfrac{1}{1+s^2}$

Since $F_c\{e^{-x}\}=\dfrac{1}{1+s^2}$ *(take $a=1$),* $f(x)=\dfrac{2}{\pi}\displaystyle\int_0^\infty \frac{\cos sx}{1+s^2}\,ds=e^{-x}$.

8. Properties of Fourier Transforms

Property	Statement $(F(s)=F\{f\})$
Linearity	$F\{af+bg\}=aF(s)+bG(s)$

Property	Statement $(F(s)=F\{f\})$
Shifting in x	$F\{f(x-a)\}=e^{isa}F(s)$
Shifting in s	$F\{e^{iax}f(x)\}=F(s+a)$
Change of scale	$F\{f(ax)\}=\frac{1}{\lvert a\rvert}F(\frac{s}{a})$
Modulation	$F\{f(x)\cos ax\}=\frac{1}{2}[F(s+a)+F(s-a)]$
Derivative	$F\{f'(x)\}=-is\,F(s),\quad F\{f''\}=-s^2F(s)$
Multiply by x	$F\{xf(x)\}=-i\frac{d}{ds}F(s)$

Cosine / sine transforms of derivatives

$$F_c\{f'(x)\}=sF_s(s)-f(0),\qquad F_s\{f'(x)\}=-sF_c(s)$$

$$F_c\{f''(x)\}=-s^2F_c(s)-f'(0),\qquad F_s\{f''(x)\}=-s^2F_s(s)+s\,f(0)$$

(valid when $f, f'\to 0$ *as* $x\to\infty$*.) Also*

$F_c\{f(x)\cos ax\}=\frac{1}{2}[F_c(s+a)+F_c(s-a)]$, *and*

$F_s\{f(x)\cos ax\}=\frac{1}{2}[F_s(s+a)+F_s(s-a)]$.

Example: $F_c\{xe^{-x}\}$ *via* $F_c\{xf\}=\frac{d}{ds}F_s\{f\}$: $\frac{d}{ds}\frac{s}{1+s^2}=\frac{1-s^2}{(1+s^2)^2}$ *(agrees with table,* $a=1$*).*

9. Convolution Theorem & Parseval's Identity

Convolution of f, g *on* $(-\infty,\infty)$: $(f*g)(x)=\int_{-\infty}^{\infty}f(u)\,g(x-u)\,du.$

Convolution & Parseval

$$F\{f*g\} = F(s)\cdot G(s) \quad \Longleftrightarrow \quad f*g = F^{-1}\{F(s)G(s)\}$$

$$\int_{-\infty}^{\infty} |f(x)|^2 dx = \frac{1}{2\pi}\int_{-\infty}^{\infty} |F(s)|^2 ds \qquad \text{(Parseval)}$$

$$\int_0^{\infty} F_c(s)G_c(s)\,ds = \frac{\pi}{2}\int_0^{\infty} f(x)g(x)\,dx$$

Example 19. Evaluate $\displaystyle\int_0^{\infty} \frac{dx}{(x^2+a^2)(x^2+b^2)}$ using Parseval

Take $f = e^{-ax}$, $g = e^{-bx}$: $F_c = \dfrac{a}{s^2+a^2}$, $G_c = \dfrac{b}{s^2+b^2}$. *Then*

$$ab\int_0^{\infty} \frac{ds}{(s^2+a^2)(s^2+b^2)} = \frac{\pi}{2}\int_0^{\infty} e^{-(a+b)x}dx = \frac{\pi}{2(a+b)}$$

$$\int_0^{\infty} \frac{dx}{(x^2+a^2)(x^2+b^2)} = \frac{\pi}{2ab(a+b)}$$

Example 20. Use Parseval to evaluate $\displaystyle\int_0^{\infty} \frac{\sin^2 x}{x^2}dx$

Pulse $f = 1$ *on* $(0,1)$: $F_c = \dfrac{\sin s}{s}$. $\displaystyle\int_0^{\infty} F_c^2 ds = \frac{\pi}{2}\int_0^{1} 1\,dx = \frac{\pi}{2} \Rightarrow$

$$\int_0^{\infty} \frac{\sin^2 s}{s^2}ds = \frac{\pi}{2}.$$

10. Quick Revision Sheet

Formulae at a glance

2π-periodic: $a_0 = \frac{1}{\pi}\int f$, $a_n = \frac{1}{\pi}\int f\cos nx$, $b_n = \frac{1}{\pi}\int f\sin nx$ (series $\frac{a_0}{2} + \ldots$).

Even f: only a_n (with $\frac{2}{\pi}\int_0^{\pi}$); odd f: only b_n. Period $2l$: replace $nx \to \frac{n\pi x}{l}$, $\frac{1}{\pi} \to \frac{1}{l}$.

Half range on $(0, l)$: $\frac{2}{l}\int_0^{l}$ with $\sin\frac{n\pi x}{l}$ (sine) or $\cos\frac{n\pi x}{l}$ (cosine).

At a jump the series gives the AVERAGE of left and right limits.

$F(s) = \int f e^{isx}dx$; $F_c = \int_0^{\infty} f\cos sx$, $F_s = \int_0^{\infty} f\sin sx$; inverses have factor $\frac{2}{\pi}$ (sine/cosine) or $\frac{1}{2\pi}$ (complex).

- Always check even/odd symmetry FIRST - it saves half the work. Quote Dirichlet conditions when asked "existence".
- Typical long questions: Fourier series of a piecewise function, half-range expansions, Fourier integral deduction ($\int \sin\lambda/\lambda$), sine/cosine transform of e^{-ax}, Parseval/convolution evaluation.